Concept of RAID-04

1. Introduction

RAID (Redundant Array of Independent/Inexpensive Disks) is a technology that combines multiple physical hard drives into a single logical unit to improve performance, provide redundancy, or both. It is widely used in servers, storage systems, and data centers to ensure data availability and reliability.

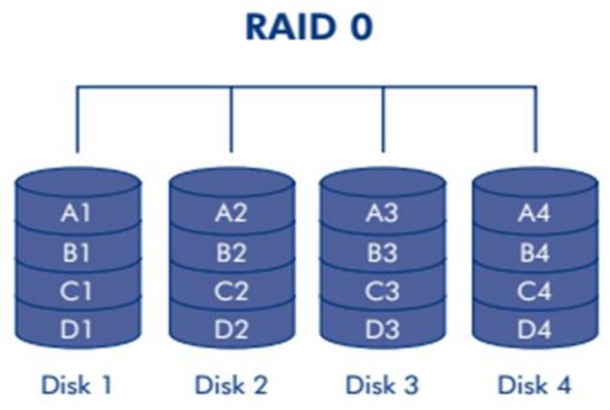
2. Objectives of RAID

- Improve read and write performance.
- Provide fault tolerance to prevent data loss.
- Combine multiple smaller drives into one large logical volume.
- Balance between performance, storage capacity, and redundancy.

3. RAID Levels

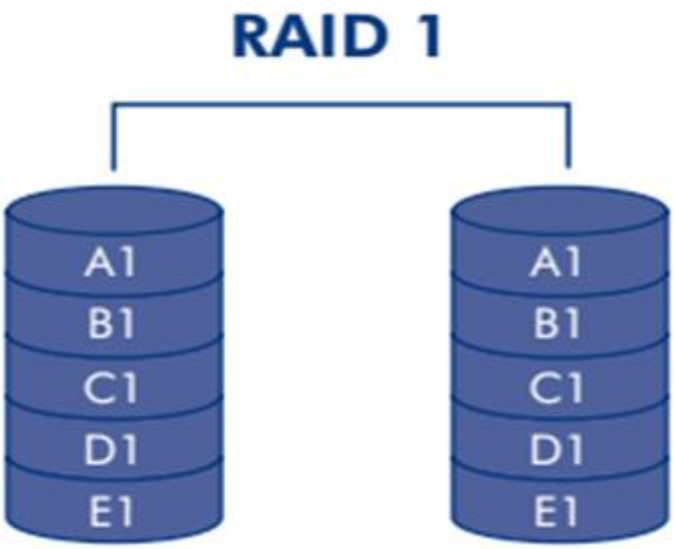
RAID 0 – Striping

- **Working**: Data is split into blocks and written across multiple drives.
- **Advantages**:
  - High performance in read/write.
  - Maximum utilization of storage capacity.
- **Limitations**:
  - No fault tolerance. If one drive fails, all data is lost.
- **Use Case**: High-speed applications like video editing or gaming



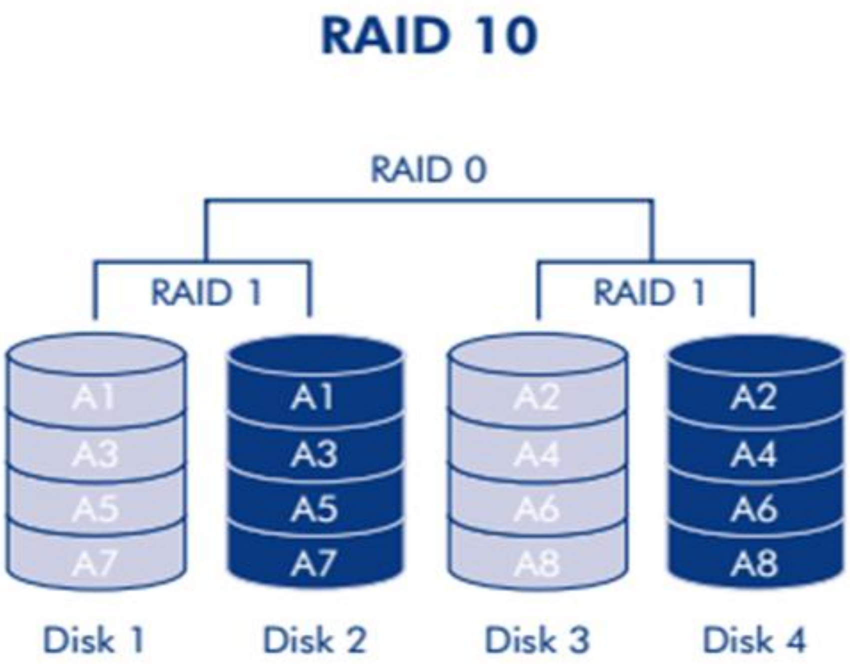
RAID 1 – Mirroring

- **Working**: Data is duplicated (mirrored) on two drives.
- **Advantages**:
    - High data reliability.
    - Simple to implement.
- **Limitations**:
    - Storage capacity is cut in half (50% usable).
- **Use Case**: Critical systems where data protection is more important than storage efficiency.



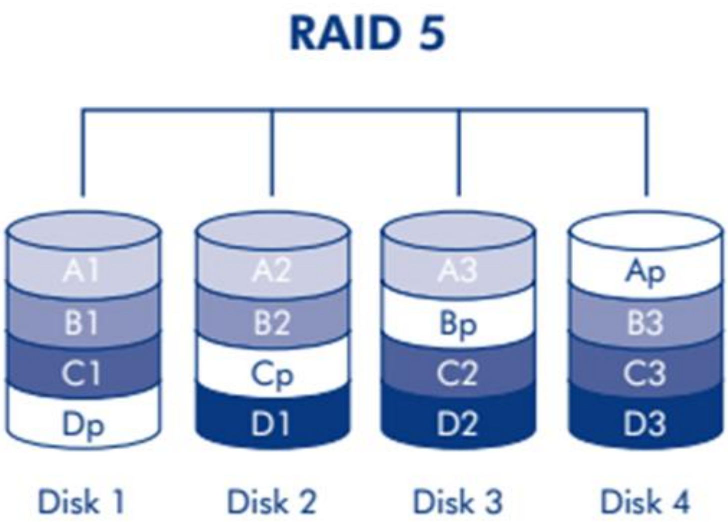
RAID 10 (1+0) – Combination of Mirroring and Striping

- **Working**: Combines RAID 1 and RAID 0. Data is mirrored first, then striped across multiple drives. Requires at least 4 drives.
- **Advantages**:
    - High performance and fault tolerance.
    - Faster recovery from failure.
- **Limitations**:
    - Expensive because 50% storage is lost to mirroring.
- **Use Case**: Databases and applications needing both speed and reliability.



RAID 5 – Striping with Parity

- **Working**: Data and parity information are striped across three or more drives. Parity helps rebuild data in case of drive failure.
- **Advantages**:
    - Balance between performance, storage efficiency, and fault tolerance.
    - Can tolerate one drive failure.
- **Limitations**:
    - Write performance is slower due to parity calculation.
- **Use Case**: File servers and general-purpose storage.



4. Advantages of RAID

- Ensures data availability.
- Improves read/write performance.
- Provides redundancy to protect against hardware failure.
- Flexible options for different needs.

5. Disadvantages of RAID

- RAID is not a replacement for backup.
- RAID controllers and additional drives increase costs.
- Some levels reduce usable storage capacity.
- Complex RAID levels require skilled management.